Healthcare Decision Intelligence Platform for Heart Failure 30-Day Readmission Risk

Iteration 4 - Scientific Report

GitHub Repository: `https://github.com/AbdulSameerS/Capstone_Healthcare_Decision_Intelligence_Agent`

Introduction

This project develops a Healthcare Decision Intelligence Platform for predicting 30-day hospital readmission risk among heart failure patients. The methodology combines structured clinical data processing, machine learning-based risk prediction, SHAP-based explainability, and a retrieval-augmented generation (RAG) pipeline with a large language model (LLM) to produce clinician-ready, patient-specific guidance. The system begins with MIMIC-IV structured data, constructs a heart failure cohort, engineers predictive features, trains and evaluates multiple machine learning models, explains the selected model through SHAP, and then combines predicted risk, top risk drivers, and patient clinical context to generate decision-support outputs such as doctor alert summaries, precautions, and follow-up recommendations.

The overall workflow of the project is: patient data → cohort construction and feature engineering → readmission risk prediction → SHAP-based explanation of top risk drivers → RAG retrieval over patient notes and clinical summaries → LLM generation of clinician-facing outputs. A system overview diagram illustrates the architecture and interaction between the main components.

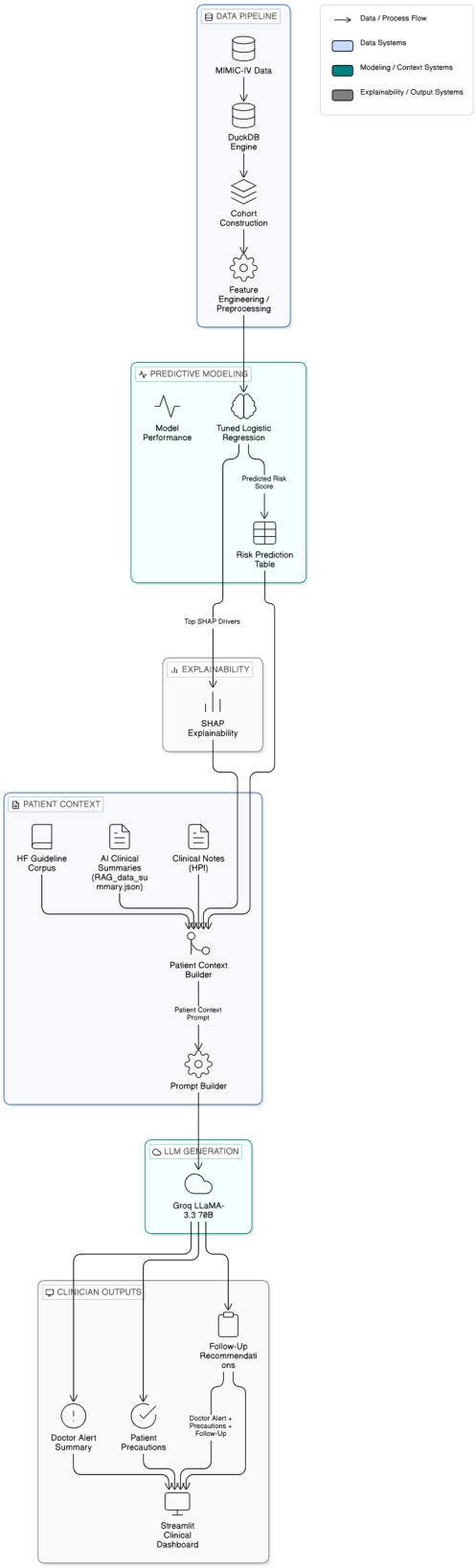

Figure 1: System overview of the Healthcare Decision Intelligence Platform

1. Purpose of the Methodology

The purpose of this methodology is to solve the problem of heart failure readmission risk prediction in a way that is not only predictive, but also interpretable and clinically useful. A risk score by itself is not sufficient in healthcare settings, where clinicians need to understand why a patient is considered high risk and what next-step guidance is relevant. For this reason, the project combines predictive modeling, SHAP-based feature-level explanation, and an LLM-powered retrieval pipeline to move from raw data to a clinician-facing decision support output.

Comparing multiple models was necessary because the dataset is imbalanced and the project required a method that performs reasonably well on the minority readmitted class while also remaining interpretable. By comparing baseline models, feature-selected models, SMOTE-based models, and tuned models, the project aimed to understand how preprocessing, feature selection, class-balancing strategies, and hyperparameter tuning affected performance. This comparison provided insight into which model was most suitable for downstream explainability and integration into the RAG and LLM pipeline.

2. Problem Statement

Defining the Problem

This project addresses a **machine learning binary classification problem**. The objective is to predict whether a heart failure patient will be readmitted within 30 days after discharge based on structured clinical, demographic, and hospitalization-related variables.

The project also includes an **LLM-based generation component**. After the readmission risk is predicted, SHAP is used to identify the most important patient-specific risk drivers. These structured explanations, together with patient clinical context and retrieved supporting information, are then used to generate a concise clinician-facing summary. Therefore, the project combines:

- Machine Learning: binary classification
- Explainable AI: feature-level interpretation using SHAP
- LLM workflow: summarization and decision-support generation

Significance

This problem is important because hospital readmissions, especially for heart failure patients, are associated with increased patient burden, poorer continuity of care, and higher healthcare costs. Better identification of high-risk patients can support more informed discharge planning, closer follow-up, and targeted precautions to reduce avoidable readmissions.

The project is relevant to academic research because it integrates structured clinical modeling, explainable AI, and generative AI in one workflow. It is relevant to healthcare and industry applications because it demonstrates how predictive models can be translated into usable decision-support outputs rather than remaining as isolated probability estimates. Existing work often stops at prediction, whereas this project extends the workflow toward explanation and actionable clinician support.

3. Data Collection and Preparation

Data Sources

The structured dataset for this project comes from the MIMIC-IV clinical database. During development, the data was handled in CSV form and then loaded into DuckDB for reproducible SQL-based analysis and feature generation. The project focuses on heart failure admissions and constructs a patient-level tabular dataset for modeling.

For the LLM and RAG portion, the project uses additional patient-context text sources, specifically unstructured clinical notes and AI-generated clinical summaries. These are represented in files such as `HPI.json` and `RAG_data_summary.json`, which are used to build patient-specific prompts for the generation pipeline.

Data Description

The final heart failure cohort contains 4,508 admissions from 4,074 unique patients after excluding deaths within 30 days. The target variable is a binary label, `readmitted_30d`, indicating whether the patient was readmitted within 30 days after discharge.

The final structured feature matrix contains 137 features after preprocessing and one-hot encoding. Feature groups include demographics, admission information, discharge-related variables, comorbidity and procedure counts, infection status, and temporal laboratory sum-

maries such as first, last, minimum, maximum, and change values for multiple lab tests. The class distribution is imbalanced, with 78.5% not readmitted and 21.5% readmitted.

For the LLM-based component, the text corpus includes patient-specific HPI notes and AI-generated summaries. These are used as contextual inputs in the retrieval and generation pipeline rather than as the main predictive dataset.

Preprocessing Steps

Several preprocessing steps were applied before modeling. First, raw MIMIC-IV data was loaded into DuckDB. Next, the heart failure cohort was constructed and the 30-day readmission label was created while excluding patients who died within 30 days to avoid leakage. Feature engineering was then used to generate structured patient-level variables from demographics, admission information, discharge information, and temporal laboratory summaries.

For machine learning preprocessing, numerical variables were handled using median imputation, while missing categorical values were represented as `UNKNOWN`. Six categorical variables were one-hot encoded, and feature scaling was applied using `StandardScaler` for Logistic Regression. Feature selection was also evaluated during model development. For the LLM-related component, patient notes and summaries were organized as structured context inputs that could later be merged with risk scores and SHAP-derived features during prompt construction.

4. Selection of Machine Learning or LLM Models

Model Consideration

The main machine learning models considered in this project were Logistic Regression and XGBoost/Gradient Boosting style models. These were selected because they provide a meaningful comparison between a simpler interpretable linear model and a more flexible non-linear model. Logistic Regression is particularly suitable for structured clinical data when interpretability is important, while Gradient Boosting can capture more complex interactions.

The project also evaluated improved versions of the models through feature selection, SMOTE-based balancing, and hyperparameter tuning. Deep learning or transformer-based models were not selected as the main predictive models because the core prediction task is based on

structured tabular clinical features, for which traditional machine learning approaches are often more suitable and easier to explain.

For the LLM portion, the model is not used as the predictive model itself. Instead, it is used after prediction and explanation to generate a structured summary using retrieved patient context. In the current implementation, the generation engine is Groq LLaMA-3.3-70B.

Final Model Selection

The final selected model for readmission prediction was **tuned Logistic Regression**. It was selected because it achieved the best balance between predictive performance and interpretability. Different models were tested to understand how baseline modeling, feature selection, class balancing, and tuning affected performance, and to determine which model would be most appropriate for SHAP-based explanation and downstream integration with the RAG pipeline.

The main evaluation metrics used for comparison were ROC-AUC, F1-score, precision, and recall. Because the dataset is imbalanced, accuracy alone was not sufficient for final selection. The final tuned Logistic Regression model achieved the strongest overall balance and was therefore chosen as the core predictive model.

5. Model Development and Training

Architecture and Configuration

The machine learning architecture is built around structured patient-level features and a binary classification model for readmission prediction. The final model uses 137 engineered features derived from structured MIMIC-IV data. The broader project architecture includes a sequence of connected components: data loading, cohort construction, feature engineering, preprocessing, model training, SHAP explainability, patient context assembly, and LLM generation.

The final predictive model was a tuned Logistic Regression model with L1 regularization and class balancing. In the LLM portion of the architecture, patient risk scores, SHAP risk drivers, clinical notes, and AI summaries are merged into a patient context that is then sent to the LLM to generate clinician-facing outputs.

Training Process

The structured dataset was split into training and test sets using a stratified 80/20 split in order to preserve class balance. The final feature set was standardized before Logistic Regression training. During experimentation, multiple versions of the model were trained, including baseline models, feature-selected models, and SMOTE-based models.

To reduce overfitting and improve reliability, cross-validation and regularization were used. In particular, stratified cross-validation was applied during hyperparameter tuning, and L1 regularization helped control model complexity. Because the project uses structured tabular data rather than transfer learning from a pretrained predictive model, transfer learning was not part of the machine learning training workflow.

Hyperparameter Tuning

Hyperparameter tuning was performed using `GridSearchCV`. Logistic Regression was tuned over regularization strength and penalty choices under a class-balanced setting. Gradient Boosting-based models were also evaluated in tuned form. Grid search was chosen because it provides a systematic and reproducible method for comparing parameter combinations and identifying the strongest configuration for the dataset.

The tuning stage led to the final Logistic Regression model, which outperformed earlier baseline versions and remained easier to interpret than more complex alternatives.

6. Evaluation and Comparison

The main metrics used to evaluate model performance were ROC-AUC, F1-score, precision, and recall. These metrics were selected because the dataset is imbalanced and the project must pay attention to performance on the minority readmitted class rather than relying only on overall accuracy.

Multiple models were evaluated, including baseline Logistic Regression, baseline XGBoost/Gradient Boosting, feature-selected versions, SMOTE-based versions, and tuned versions. This comparison showed that Logistic Regression consistently remained competitive and became the strongest final model after tuning. According to the current project repository, the final tuned Logistic Regression model achieved a test ROC-AUC of 0.6546. SHAP was then applied to this final model to generate both global and patient-level explanations.

For the LLM portion, formal human evaluation metrics such as ROUGE or perplexity were not the primary focus in this iteration. Instead, the emphasis was on building the generation pipeline that transforms structured patient context into doctor alerts, precautions, and follow-up recommendations. These outputs are intended to be evaluated more qualitatively in the context of clinical usefulness and grounding.